

Highlights

A Network-Level Parental Control System Integrated with Home Wi-Fi Routers and Cloud Services: Design, Performance Evaluation and Technology Acceptance

Anonymised for review

- Network-level parental control with router–cloud–mobile integration on real hardware.
- Filtering rules applied with mean latency below 800 ms; throughput degradation under 5%.
- TAM evaluation with 110 caregivers shows high acceptance (Cronbach's $\alpha = 0.89$).
- Significantly higher perceived usefulness and attitude than Google Family Link.
- Less-intrusive supervision avoids on-device content inspection and risky permissions.

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Abstract

The intensive use of the Internet by children and adolescents in domestic settings has outpaced the capacity of conventional, device-bound parental control applications, which are easy to bypass, fragmented across devices, and frequently perceived as overly intrusive by minors. This paper proposes, implements and validates a network-level parental control system that integrates an Android mobile application with a domestic Wi-Fi router (TP-Link Archer C6 V3.2) and a cloud-based DNS filtering service (NextDNS) synchronised through Firebase. Requirements were elicited through a PRISMA-compliant systematic literature review of fifteen peer-reviewed studies (Cohen's $\kappa = 0.783$) and a competitive analysis of four widely-used parental control applications. The prototype was developed under the Rapid Application Development methodology and evaluated in a real domestic Wi-Fi environment along three complementary axes: (i) network performance, measured through rule-application latency, application-network synchronisation time, cloud-resource consumption and `iperf3` throughput tests; (ii) technology acceptance, assessed through the Technology Acceptance Model (TAM) with 110 caregivers using an 11-item Likert questionnaire; and (iii) a within-subjects comparison against Google Family Link using the Wilcoxon signed-rank test with Bonferroni correction. The system applied filtering rules with an average latency of 398–734 ms across 1–10 connected devices, synchronised events to the parental interface in 2.6 s on average, and degraded throughput by less than 5% in all scenarios. The TAM evaluation produced high acceptance scores in perceived usefulness ($M=4.29$), perceived ease of use ($M=4.17$) and attitude towards use ($M=4.36$), with Cronbach's $\alpha = 0.89$ and a confirmatory factor analysis exhibiting an excellent fit (CFI=0.961, TLI=0.953, RMSEA=0.063, SRMR=0.049). The proposed system significantly outperformed Google Family Link on perceived usefulness ($p = 0.012$, $r = 0.41$) and attitude ($p = 0.007$, $r = 0.46$). The findings show that shifting parental supervision from the device to the home-network layer can preserve technical effectiveness while reducing the privacy intrusiveness that has eroded acceptance of traditional parental control applications.

Keywords: Parental control, Home network, DNS filtering, Mobile application, Technology Acceptance Model, Children online safety

1. Introduction

The diffusion of always-on connectivity, smartphones and connected devices in family homes has moved children’s media consumption from a single, supervisable screen to a dispersed ecosystem of personal devices that parents struggle to oversee (Holmgren et al., 2024; Liu et al., 2025). Recent panel and survey work confirms that adolescents’ problematic smartphone use is rising and is consistently associated with the depth and quality of parental mediation strategies, with parental monitoring acting as a significant protective factor whose effectiveness is moderated by self-control and the home digital ecology (Ho et al., 2025; Liu et al., 2025; Sevilla-Fernández et al., 2025). At the same time, the predominantly device-centred parental control market is repeatedly criticised in human–computer–interaction and behavioural research for over-restricting adolescents, eroding parent–child trust and producing unintended security and privacy harms (Wang et al., 2021; Akter et al., 2022; Feal et al., 2020; Akter et al., 2025).

This tension between effective protection and respectful mediation is particularly relevant for *Computers in Human Behavior*, where work over the past two years has converged on three findings. First, restrictive mediation reduces exposure to online risks but also reduces digital opportunities, while active mediation requires technical competence that many caregivers do not feel they possess (Holmgren et al., 2024; Sciacca et al., 2022). Second, parents’ self-efficacy and awareness of online risks are imperfectly correlated, so confident parents are not necessarily aware parents (Kuldas et al., 2024). Third, the digital harms of smart-home devices—including invasive data collection by parental control apps themselves—are now recognised as a behavioural and design problem, not only a technical one (Buil-Gil et al., 2023; Feal et al., 2020). These findings push researchers to look for parental control architectures that are technically robust, respectful of family privacy, and easy enough to be adopted by non-expert caregivers.

A complementary observation comes from the technical literature: most popular parental control applications operate on the child’s device, which makes them easy to evade by uninstallation, factory reset, mobile-data switching, or VPN tunnelling, and forces them to demand intrusive permissions (Feal et al., 2020; Canino et al., 2025; Wang et al., 2021). By contrast, controls implemented on the home Wi-Fi router or via Programmable DNS (PDNS) services govern *all* devices that traverse the household network and do not require installing software on each child’s device. Recent technical evaluations show, however, that current consumer PDNS and router-level tools differ widely in coverage, blocking precision and resistance to evasion (Canino et al., 2025; Patel et al., 2024). There is therefore an opportunity to design a network-level parental control architecture that combines (i) a router-anchored enforcement point, (ii) a cloud DNS filtering layer with personalised rules, and (iii) a mobile application that is the family’s single locus of supervision and that has been validated against real users.

The present work addresses this opportunity. We design, implement and evaluate RED-PARENTAL, an Android parental control application that is integrated, on real hardware, with a domestic TP-Link router and with a NextDNS-based filtering service synchronised through Firebase Cloud Functions. The system was elicited from a PRISMA-compliant systematic literature review and a competitive analysis of four widely deployed applications, and was evaluated end-to-end on three layers: (a) technical performance in a controlled domestic Wi-Fi setting; (b) acceptance with 110 caregivers using a Technology Acceptance Model (TAM) instrument with formal psychometric validation; and (c) a within-subjects comparison against Google Family Link.

The contributions of this paper are:

1. A network-level parental control architecture that integrates a mobile application, a domestic Wi-Fi router and a cloud DNS service, with verified interoperability against real hardware (TP-Link Archer C6 V3.2) and reproducible reverse-engineered authentication.
2. A multi-metric technical evaluation in a real Wi-Fi environment covering rule-application latency, end-to-end synchronisation time, Firestore resource usage under increasing device load, and `iperf3` throughput degradation.
3. A TAM-based behavioural evaluation with 110 caregivers including reliability (Cronbach’s α), confirmatory factor analysis, convergent and discriminant validity, and a Wilcoxon-based comparison with Bonferroni correction against Google Family Link.
4. A discussion of how a network-level, less-intrusive design changes the trade-off between protection and surveillance highlighted in recent CHB literature ([Wang et al., 2021](#); [Akter et al., 2022](#); [Sevilla-Fernández et al., 2025](#)).

The remainder of the paper is organised as follows. Section 2 reviews JCR-indexed work on parental mediation, parental control applications and network-based supervision. Section 3 describes the materials and methods, including the systematic review, the development methodology and the evaluation design. Section 4 reports the technical and TAM results, while Section 5 discusses them against the literature. Section 6 concludes and Section 7 outlines future work.

2. Related work

2.1. Parental mediation: from device-bound restriction to network-level governance

Parental mediation has been studied for two decades along three axes: *active* mediation (talking with the child about content), *restrictive* mediation (banning, blocking or limiting access), and *co-use* (consuming media together) ([Sciacca et al., 2022](#); [Holmgren et al., 2024](#)). Recent JCR-indexed studies converge on three points relevant for the design of parental control technologies. First, restrictive mediation reduces online risks but also

reduces opportunities and digital skills, while active mediation broadens both (Sevilla-Fernández et al., 2025; Liu et al., 2025). Second, parents’ self-efficacy is not always aligned with their actual awareness of online risks, so technical tools are needed to bridge that gap (Kuldass et al., 2024). Third, the maternal/paternal media-use profile shapes the household’s digital culture and the effectiveness of any control tool deployed in it (Holmgren et al., 2024). In a Chinese sample of 600 families, Liu et al. (2025) showed that authoritative parenting is linked to lower internet addiction levels and that active parental mediation partially mediates this relationship.

2.2. Parental control applications: design space and trade-offs

The design space of parental control apps was systematically mapped by Wang et al. (2021), who analysed 58 mobile apps and 3,206 reviews and found that the prevailing market designs are restrictive and surveillance-oriented, generating negative reviews from teenagers who perceive them as punishment rather than protection. Akter et al. (2022) extended this critique by proposing a joint family-oversight model in which parents and adolescents share supervision responsibilities. The HCI community has more recently called for “community-based” approaches to adolescent online safety that move beyond individual parental surveillance toward broader support networks (Akter et al., 2025). From the privacy and security side, Feal et al. (2020) audited 46 Android parental control apps and reported that a substantial portion exhibit invasive permissions, third-party tracking and weak protection of the data they collect. These findings motivate designs that minimise data collection on the child’s device.

2.3. Network-level and DNS-based parental control

A complementary line of research locates the enforcement point at the home network rather than at the device. Canino et al. (2025) performed a technical evaluation of Android parental control tools and PDNS filtering services in the Chilean context, testing 1,427 websites; they reported wide variability in blocking precision and recall, with some commercial PDNS services failing to block more than half of adult content. Patel et al. (2024) proposed a granular network-traffic classification approach that uses recurrent neural networks to identify mobile applications even when traffic is encrypted, achieving up to 80% accuracy and enabling categorical control beyond URL blocking. These technical works confirm that network-level filtering is feasible but that it must be carefully designed to handle modern encrypted traffic.

2.4. Mobile parental control prototypes (2020–2025)

Several prototypes have been proposed in recent years. TEEN-ALYSE (Mahanani et al., 2021) couples parental control with teen self-monitoring and active mediation through a YouTube content classifier (80.25% precision over 400 links). CHILDSAFE (Magpantay and Albano, 2025) is an offline Android system that manages screen time and app restrictions

with a PIN-based parental console. KINDERGUARD (Pradeep et al., 2024) uses face recognition (LBPH) to switch the device automatically into a child mode. GUARDIAN EYE (Tashan and Ramachandran, 2024) integrates a proxy server with a mobile app and a web panel for content filtering. KEYAWARE (Adekunle et al., 2025) explores ethical keyloggers for child online safety, while Ravichandran and Suganya (2024) present a geofence-based teen-location monitor. The Chilean technical evaluation by Canino et al. (2025) and the comparative study by Ahmad and Khan (2025) illustrate the heterogeneity of the design space. Studies focused on family dynamics (Wang et al., 2021; Akter et al., 2022; Feal et al., 2020) stress that the technical design has direct behavioural consequences: punitive or invasive designs reduce acceptance and trust between parents and adolescents.

2.5. Research gap

Three gaps emerge from this literature. (G1) Most prototypes are evaluated either in technical conditions or in surveys of caregivers, but rarely in both. (G2) Network-level designs are technically promising but have been evaluated against limited numbers of websites or in laboratory conditions, without behavioural acceptance evidence. (G3) Few studies report a quantitative comparison against a widely deployed commercial baseline using formal psychometric instruments. The present work addresses these three gaps simultaneously.

3. Materials and methods

3.1. Overview

A non-experimental, cross-sectional design was adopted, organised in four phases (Figure 1): (1) systematic literature review and competitive analysis to elicit requirements; (2) iterative software construction following the Rapid Application Development (RAD) methodology (Martin, 1991); (3) technical evaluation in a real domestic Wi-Fi environment; and (4) behavioural evaluation through a TAM survey and a within-subjects comparison with Google Family Link. The Quevedo State Technical University Ethics Committee approved the protocol, and all participants provided informed consent.

3.2. Systematic literature review

The review followed the PRISMA framework (Page et al., 2021) and the PICOC strategy. The search string combined terms on *parental control*, *home network*, *DNS/content filtering* and *children/adolescents*, and was executed in May 2026 over six databases (IEEE Xplore, ACM DL, Scopus, Taylor & Francis, ScienceDirect and Web of Science). Inclusion criteria were peer-reviewed studies in English, 2020–2026, with empirical evidence of implementation, evaluation or analysis of parental control applications. Exclusion criteria covered posters, work-in-progress, editorials and abstracts. Two reviewers screened the studies independently; the inter-rater agreement (Cohen’s $\kappa = 0.783$) indicated substantial concordance (Landis and Koch, 1977). From 757 records, 15 were retained.

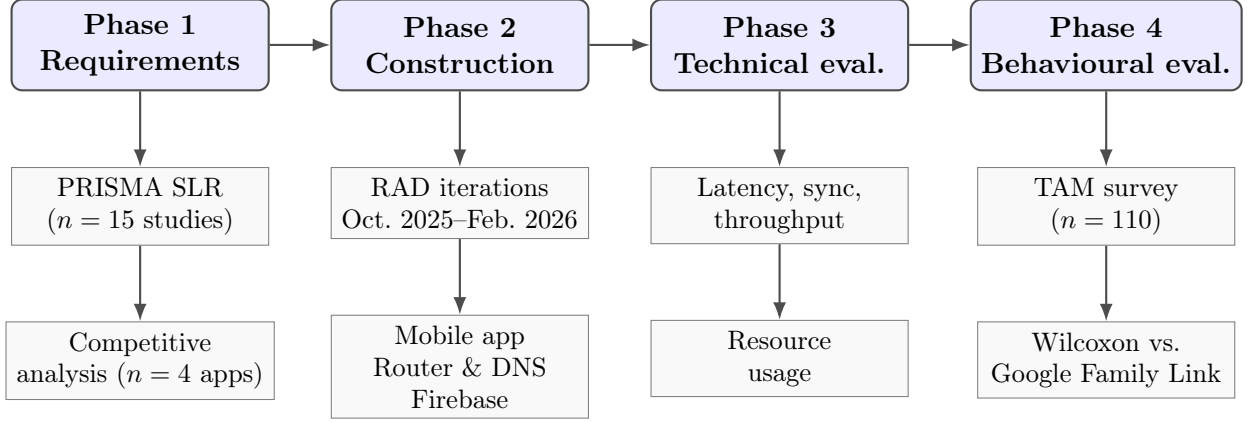


Figure 1. Overall research workflow.

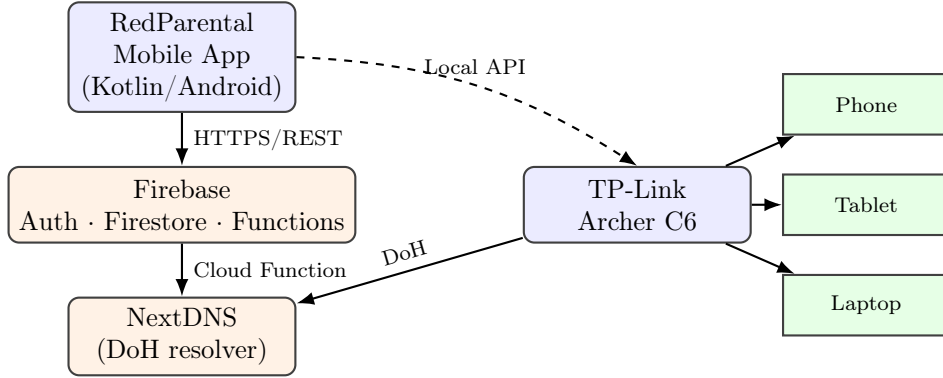


Figure 2. High-level system architecture: a Kotlin/Android client communicates with Firebase, which propagates filtering rules to NextDNS; the home router uses DNS-over-HTTPS so all devices on the household Wi-Fi inherit the rules.

3.3. System design and implementation

3.3.1. Architecture

The system follows a three-layer architecture (Figure 2). The *client layer* is an Android (Kotlin) mobile application that authenticates the caregiver and exposes a single dashboard for the household. The *network layer* comprises a TP-Link Archer C6 V3.2 router (IEEE 802.11ac, dual band, WPA2/WPA3) and the NextDNS resolver, configured with DNS-over-HTTPS (DoH). The *cloud layer* relies on Firebase (Authentication, Firestore, Cloud Functions) for identity, persistence and rule synchronisation between the mobile app and the DNS resolver. Communication between layers uses HTTPS and a REST API; rule deployment to NextDNS is triggered by Firestore writes that activate Cloud Functions.

3.3.2. Router integration

To enable direct communication with the router, the panel’s authentication flow was reverse-engineered following the steps proposed in API discovery work (Liu et al., 2024). Network traffic between a browser and the panel was captured using the developer tools and exported in HAR format; the request sequence was replayed in Postman to identify endpoints and parameters; and the public `tpEncrypt.new.js` file was inspected, revealing

an XOR pre-obfuscation step, an RSA key exchange and an AES session cipher (Stallings, 2017). A Kotlin module (TPLinkHelper) and a cryptographic helper (TPLinkCrypto) replicate the protocol over an OkHttp client, enabling authenticated extraction of the connected-device list and per-device control actions.

3.3.3. Development methodology

The application was built between October 2025 and February 2026 in 1–2 week iterations under the RAD methodology (Martin, 1991). The IEEE 29148:2018 standard (ISO/IEC/IEEE, 2018) guided requirement elicitation, yielding 19 functional and 10 non-functional requirements (the most distinctive being RNF-09 *interoperability* with the home router). Source code is available at the project repository.

3.4. Technical evaluation

The system was evaluated on a residential 100/20 Mbps fibre connection with all devices on the same IEEE 802.11ac Wi-Fi network. Time was synchronised across devices via NTP (time.google.com). Three test devices were used (Galaxy A24, Redmi Note 11 and a development laptop). The following metrics were collected:

- **Rule-application latency.** Time elapsed between activating a blocking rule from the mobile app and the router stopping responses to the blocked domain (e.g. www.youtube.com), measured by ping, repeated for 1, 5 and 10 connected devices.
- **End-to-end synchronisation.** Time between a navigation event on the network and its appearance in the mobile dashboard, measured over 50 events.
- **Cloud resource usage.** Cloud Function invocations per minute, execution time, allocated memory and Firestore reads/writes during 30-minute sessions, recorded from the Firebase console with 1, 5 and 10 devices.
- **Throughput impact.** iperf3 runs (30 s, 10 repetitions) with and without the system enabled, computing degradation as $\Delta = (V_{\text{off}} - V_{\text{on}})/V_{\text{off}} \times 100\%$.

3.5. Behavioural evaluation: TAM

A TAM (Davis, 1989) questionnaire was deployed in Google Forms to 110 caregivers recruited through convenience and snowball sampling. The instrument contained 11 items measured on a 5-point Likert scale, distributed across three constructs: *perceived usefulness* (PU, 5 items), *perceived ease of use* (PEOU, 4 items) and *attitude towards use* (ATU, 2 items). A 20-participant pilot ensured clarity. The sample size was justified *a priori* in G*Power 3.1.9.7: a single-mean *t*-test required $n = 27$ ($\alpha = 0.05$, $1 - \beta = 0.80$, $d = 0.5$), and a multiple linear regression with two predictors and five total variables required $n = 107$ ($\alpha = 0.05$, $1 - \beta = 0.95$, $f^2 = 0.15$). The final sample of 110 exceeds both thresholds.

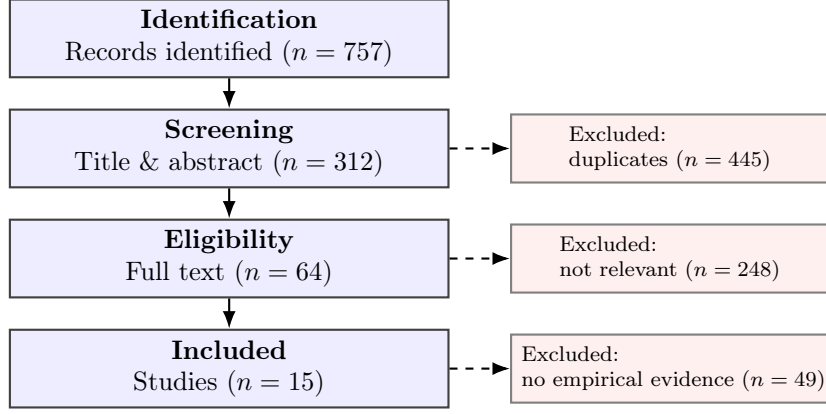


Figure 3. PRISMA flow diagram of the systematic review.

Reliability was assessed by Cronbach’s α , validity by confirmatory factor analysis (CFA) using the lavaan package in R (Rosseel, 2012), with model fit indices CFI, TLI, RMSEA and SRMR (Hu and Bentler, 1999), and the Average Variance Extracted (AVE) and the Fornell–Larcker criterion (Fornell and Larcker, 1981) for convergent and discriminant validity. Categorical associations were tested with χ^2 .

3.6. Comparison with Google Family Link

The same 110 participants evaluated both REDPARENTAL and Google Family Link in a partially counterbalanced within-subjects design with at least 24 h between sessions. The Wilcoxon signed-rank test (Wilcoxon, 1945) was applied to each TAM dimension; the effect size was computed as $r = Z/\sqrt{N}$, with $r > 0.3$ interpreted as moderate (Cohen, 1988). Because three dimensions were tested simultaneously, the significance level was adjusted with Bonferroni correction to $\alpha_{\text{corrected}} = 0.05/3 = 0.0167$.

3.7. Privacy and ethics

The system processes only network-related data (MAC addresses, hostnames, DNS queries, timestamps). No content of communications is inspected. Data are stored encrypted at rest in Firebase, with role-based access control. Participants’ identities were anonymised before analysis. Caregivers could withdraw at any time and request deletion of their data.

4. Results

4.1. Systematic literature review

The PRISMA process retained 15 studies from 757 records (Figure 3). Table 1 summarises the included studies, which span four design families: (i) mobile-only restrictive apps, (ii) AI-augmented prototypes, (iii) mediation-oriented designs and (iv) network/DNS-level evaluations. Restrictive on-device apps remain dominant, while network- and DNS-level evaluations are scarce and recent.

Table 1. Selected studies in the systematic review.

Ref.	Focus	Year
(Mahanani et al., 2021)	Active mediation, YouTube classifier	2021
(Magpantay and Albano, 2025)	Offline Android, screen time	2025
(Pradeep et al., 2024)	AI/face-recognition child mode	2024
(Tashan and Ramachandran, 2024)	Proxy server + web panel	2024
(Mounesh et al., 2022)	AI-based parental control	2022
(Wijesinghe et al., 2023)	Digital monitoring & control	2023
(Khan and Sharma, 2025)	AI mobile usage tracking	2025
(Adekunle et al., 2025)	Ethical keylogger	2025
(Ravichandran and Suganya, 2024)	Geofence teen location	2024
(Patel et al., 2024)	Granular traffic classification	2024
(Canino et al., 2025)	Android & PDNS evaluation (Chile)	2025
(Ahmad and Khan, 2025)	Keyloggers vs. traditional	2025
(Wang et al., 2021)	Design space, protection vs. punishment	2021
(Akter et al., 2022)	Joint family oversight	2022
(Feal et al., 2020)	Privacy audit of 46 apps	2020

Table 2. Comparison of representative parental control applications.

Feature	Tether	Qust.	AirD.	Fam.	Link	Ours
Router-level enforcement	✓	—	—	—	—	✓
Cloud DNS filtering	—	—	—	—	—	✓
Per-child profile	—	✓	✓	✓	✓	✓
Real-time activity dashboard	—	✓	✓	✓	✓	✓
Time-based rules	✓	✓	✓	✓	✓	✓
No on-device app required	✓	—	—	—	—	✓

Table 3. Filtering rule application latency across load scenarios.

Scenario	Mean (ms)	SD	P95 (ms)	P99 (ms)
1 device	398	52	510	590
5 devices	562	68	680	790
10 devices	734	81	890	1020

4.2. Functional comparison with market apps

A competitive analysis of TP-LINK TETHER, QUSTODIO, AIRDROID PARENTAL CONTROL and GOOGLE FAMILY LINK is summarised in Table 2. The proposed system is the only one that combines router-level enforcement, cloud DNS filtering and a single mobile dashboard.

4.3. Technical performance

4.3.1. Rule-application latency

The mean latency between activating a rule and the router blocking the domain ranged from 398 ms (1 device) to 734 ms (10 devices), with the 99th percentile staying below 1.1 s in the heaviest scenario (Table 3, Figure 4). All values are below the 5 s functional threshold defined in RNF-06.

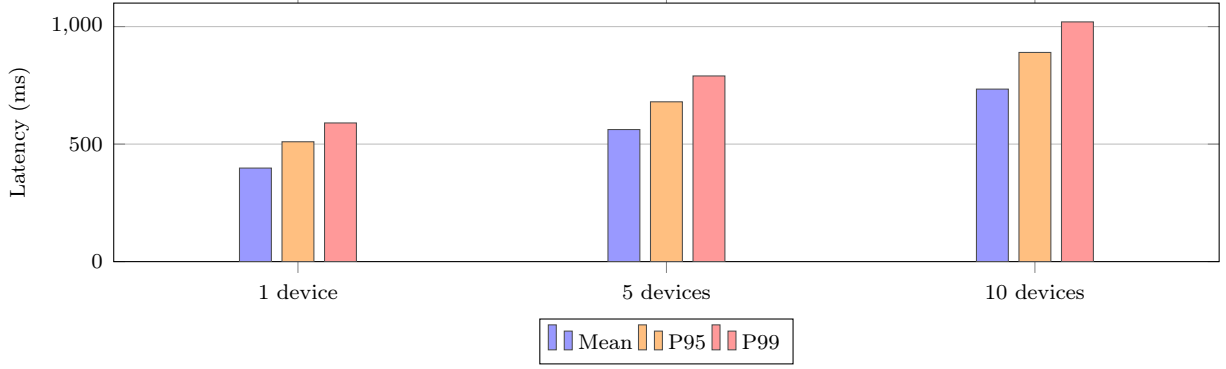


Figure 4. Filtering rule application latency: mean, P95 and P99 by number of connected devices.

Table 4. Cloud-side resource usage during 30-min sessions.

#Devices	Inv./min	Time (ms)	Mem (MB)	Reads	Writes
1	8	120	128	45	12
5	26	145	160	130	38
10	52	178	192	260	74

Table 5. Throughput with and without the system enabled (`iperf3`).

Scenario	Off (Mbps)	On (Mbps)	Degradation
1 device	96	94	2.1%
5 devices	94	91	3.2%
10 devices	92	88	4.3%

4.3.2. End-to-end synchronisation

Across 50 events, the application surface reflected a navigation event in 2.6 ± 0.7 s on average, with P95 of 4.1 s. This is well within the 5 s threshold of RNF-06, supporting the perception of near-real-time supervision.

4.3.3. Cloud resource usage

Resource consumption scaled near-linearly with the number of connected devices (Table 4). Memory remained below 200 MB and execution time below 200 ms even with 10 devices.

4.3.4. Throughput impact

Throughput degradation was below 5% in all scenarios (Table 5, Figure 5), confirming that DoH-based filtering does not noticeably affect ordinary domestic broadband performance.

4.4. Sample

The sample (Table 6) was balanced by gender (51.8% male, 48.2% female), with a majority of participants aged 20–39 (66.4%). Mobile experience was distributed across basic (30.9%), intermediate (34.5%) and advanced (34.5%). 52.7% had previously used a parental control app.

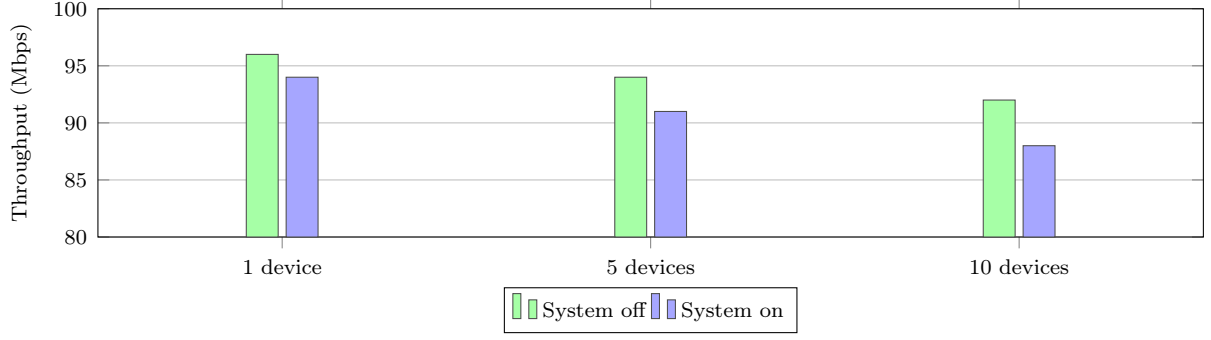


Figure 5. Throughput with and without the system, by number of connected devices.

Table 6. Demographic characteristics of the participants ($n = 110$).

Variable	Category	n	%
Gender	Male	57	51.8
	Female	53	48.2
Age	20–39	73	66.4
	≥ 40	37	33.6
Mobile exp.	Basic	34	30.9
	Intermediate	38	34.5
	Advanced	38	34.5
Children	1	38	34.5
	2	35	31.8
	≥ 3	37	33.6
Prior PC app use	Yes	58	52.7
	No	52	47.3
Daily Internet use	3–7 h	53	48.2
	> 7 h	57	51.8

Table 7. Reliability of the TAM instrument.

Dimension	Items	α	Interpretation
PU	5	0.87	High
PEOU	4	0.85	High
ATU	2	0.83	Good
Total	11	0.89	Excellent

4.5. Reliability and validity of the TAM instrument

Cronbach’s α exceeded 0.80 for every construct and reached 0.89 for the full scale (Table 7). The CFA showed an excellent fit: CFI=0.961, TLI=0.953, RMSEA=0.063, SRMR=0.049 (Hu and Bentler, 1999) (Table 8). All AVE values exceeded 0.50 and the Fornell–Larcker criterion was satisfied (Table 9).

4.6. TAM scores

The three constructs received high scores (Table 10, Figure 6). Attitude towards use ($M = 4.36$) was the highest, followed by perceived usefulness ($M = 4.29$) and perceived ease of use ($M = 4.17$).

Table 8. CFA fit indices.

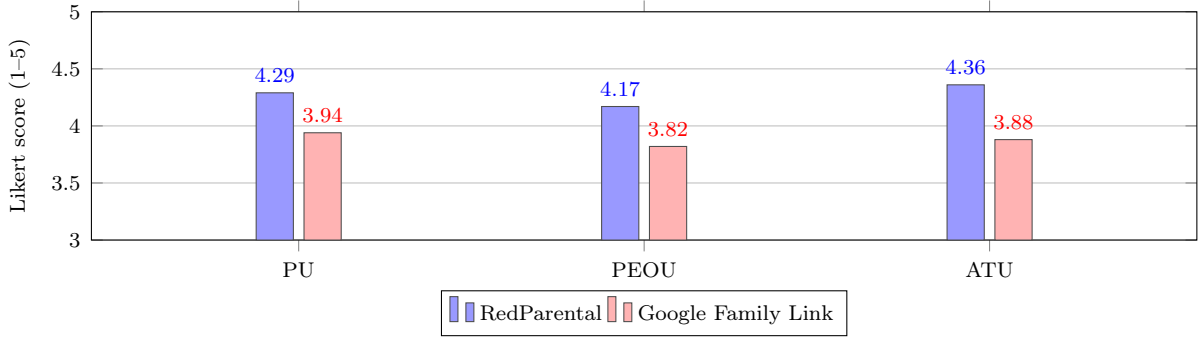
Index	Obtained	Threshold	Result
CFI	0.961	> 0.95	Excellent
TLI	0.953	> 0.95	Adequate
RMSEA	0.063	< 0.08	Acceptable
SRMR	0.049	< 0.08	Adequate

Table 9. Convergent and discriminant validity (Fornell–Larcker).

Dim.	AVE	$\sqrt{AVE}$	PU	PEOU	ATU
PU	0.69	0.83	0.83		
PEOU	0.66	0.81	0.59	0.81	
ATU	0.72	0.85	0.57	0.61	0.85

Table 10. TAM scores ($n = 110$).

Dimension	Mean	SD	95% CI
PU	4.29	0.61	[4.18, 4.40]
PEOU	4.17	0.64	[4.05, 4.29]
ATU	4.36	0.58	[4.25, 4.47]

**Figure 6.** TAM scores: REDPARENTAL vs. Google Family Link across the three dimensions.**Table 11.** Within-subjects comparison via Wilcoxon signed-rank test (Bonferroni-corrected $\alpha = 0.0167$).

Dim.	Ours	Family Link	p	r
PU	4.29	3.94	0.012*	0.41
PEOU	4.17	3.82	0.019	0.38
ATU	4.36	3.88	0.007*	0.46

* Significant after Bonferroni correction.

4.7. Comparison against Google Family Link

The Wilcoxon signed-rank test produced moderate effect sizes in all three dimensions (Table 11). After Bonferroni correction at $\alpha = 0.0167$, perceived usefulness ($p = 0.012$, $r = 0.41$) and attitude towards use ($p = 0.007$, $r = 0.46$) remained significant, while perceived ease of use ($p = 0.019$, $r = 0.38$) did not pass the corrected threshold.

Table 12. Themes from open-ended responses ($n = 110$).

Theme	Freq.	%
App and website blocking	63	57.3
Real-time device monitoring	40	36.4
Easy and intuitive interface	22	20.0
Activity notifications and alerts	17	15.5
Personalisation of supervision	14	12.7

4.8. Qualitative responses

A thematic analysis of the open-ended responses produced five themes (Table 12). Filtering of apps and websites and real-time monitoring were the most frequently valued features.

5. Discussion

5.1. The network as the right control surface

The technical results indicate that a network-level architecture can support the supervision needs of an average household with low overhead. Filtering rules took at most 734 ms to take effect with 10 connected devices, the synchronisation between events on the network and the mobile dashboard remained around 2.6 s, and throughput degradation was below 5% in every scenario. These figures address one of the recurring objections in the literature on PDNS-based filtering, namely that it can degrade user experience and produce unpredictable behaviour (Canino et al., 2025). The architecture’s reliance on Firebase for synchronisation and on NextDNS for resolution made the household’s enforcement layer independent of the child’s device, which mitigates the well-documented evasion patterns of on-device parental control apps (Feal et al., 2020; Wang et al., 2021).

5.2. Acceptance and the privacy-protection trade-off

The TAM results—PU=4.29, PEOU=4.17, ATU=4.36, with $\alpha = 0.89$ and a CFA fit consistent with the cutoffs of Hu and Bentler (1999)—indicate that caregivers found the system useful, easy to use and worth recommending. The Wilcoxon comparison against Google Family Link showed significant advantages in perceived usefulness and attitude after Bonferroni correction, with effect sizes that the literature considers moderate (Cohen, 1988). Importantly, the dimension that did not survive the Bonferroni threshold was perceived ease of use, which is the construct most sensitive to participants’ technical familiarity. This finding is consistent with the conclusions of Holmgren et al. (2024) and Kuldass et al. (2024): differences in caregivers’ digital self-efficacy can mask differences in the actual usability of a system. The qualitative themes confirmed the priority of filtering and real-time monitoring, and pointed to alerts and personalisation as the most desired enhancements.

The most important behavioural takeaway is that a less-intrusive design—network-level filtering instead of on-device content inspection—received *higher* acceptance scores than a major commercial baseline. This contradicts the implicit assumption of much of the parental control market that more visibility on the child’s device equates with more parental satisfaction (Wang et al., 2021; Akter et al., 2025; Feal et al., 2020).

5.3. Comparison with the literature

The latency and synchronisation figures are consistent with the technical pattern reported by Canino et al. (2025), with the additional advantage of integrating real-time synchronisation through Firebase. Compared with on-device prototypes such as TEEN-ALYSE (Mahanani et al., 2021), CHILDSAFE (Magpantay and Albano, 2025) or KINDERGUARD (Pradeep et al., 2024), the proposed system avoids inspecting on-device content—a practice that has produced privacy concerns in audits such as Feal et al. (2020). The behavioural acceptance results align with Akter et al. (2022), who argued that joint, less-surveillant designs produce higher acceptance, and with the design-space critique by Wang et al. (2021).

5.4. Limitations

Three limitations should be acknowledged. First, technical evaluation used a single router model (TP-Link Archer C6 V3.2); generalisation to other hardware would require replication. Second, the TAM evaluation was cross-sectional, capturing acceptance at a single point in time rather than after sustained use. Third, the sample, although powered, was a convenience sample biased towards participants with intermediate-to-advanced mobile experience, which may inflate PEOU. None of these limitations affects the validity of the relative comparison with Google Family Link, which used the same participants in a within-subjects design.

6. Conclusion

This paper introduced REDPARENTAL, a network-level parental control system that integrates a Kotlin/Android client, a TP-Link domestic router and a NextDNS-based DoH filtering service synchronised through Firebase. The architecture, elicited from a PRISMA-compliant systematic review of fifteen peer-reviewed studies and a competitive analysis of four widely deployed applications, was evaluated end-to-end in a real domestic Wi-Fi setting. The system exhibited stable performance (mean rule-application latency between 398 and 734 ms, end-to-end synchronisation of 2.6 s, throughput degradation below 5%), high reliability and validity in a TAM-based evaluation with 110 caregivers (Cronbach’s $\alpha = 0.89$; CFI=0.961, RMSEA=0.063), and significantly higher perceived usefulness and attitude than Google Family Link in a within-subjects Wilcoxon comparison with Bonferroni correction. The findings support the broader claim emerging in recent CHB literature:

parental control systems that move enforcement away from the child’s device toward the household network can be both technically effective and behaviourally well-accepted, while reducing the privacy intrusiveness that has eroded trust in conventional parental control apps.

7. Future work

Four lines of future work follow. First, we plan to replicate the technical evaluation across multiple router fabricants and firmware versions to characterise the boundaries of interoperability. Second, we will conduct a longitudinal pre–post study to evaluate sustained changes in caregivers’ perceived control, parent–teen trust and confidence in digital parenting. Third, we will extend the comparison against commercial baselines to include device-side and PDNS-only solutions under controlled traffic mixes, using the metrics defined in [Canino et al. \(2025\)](#). Finally, we will explore extensions to AI-driven content classification for traffic that bypasses DNS resolution (e.g. VPN, encrypted DoT/DoH from end devices) along the lines of [Patel et al. \(2024\)](#), while keeping the privacy-by-design principle of avoiding on-device content inspection.

CRedit authorship contribution statement

[Anonymised for review.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Anonymised data underlying the TAM evaluation will be made available by the corresponding author upon reasonable request.

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